EXCERPT — PP. 328–329

On the Divalent Carbon Atom. [Fourth Treatise.] The Chemistry of Methylene.

J. U. Nef

[Communication from the Kent Chemical Laboratory of the University of Chicago, U.S.A.]

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*Translated excerpt: the passage on pp. 328–329 dealing with the combustion of ether and Legler's peroxide. It begins and ends mid-discussion within the larger memoir (Liebigs Ann. **298**, 202–374).*

… was demonstrated by its conversion through potash into the green-black, explosive precipitate. The experiments in this direction are being continued and will then be published in detail by Mr. Neher.

If one further takes into account the fact that a mercury salt of vinyl alcohol has been isolated having entirely different properties, and that acetaldehyde, according to Polleck and Thümmel — as I have confirmed — gives no precipitate with their mixture (a solution of corrosive sublimate in potassium bicarbonate), then one may conclude with certainty that vinyl alcohol is not present in impure ether. That the impurity consists of divinyl ether is already to be regarded as probable.

Now, in the combustion of ether at higher temperature (260–500°), or in the presence of a heated platinum spiral, as Legler has shown, there arise not only hydrogen peroxide, aldehyde, and acetic acid, but also formaldehyde and formic acid. These differences are explained by the fact that at higher temperature a dissociation of the ethylene formed by decomposition of ether[198] into two molecules of methylene sets in; the methylene then burns to hydrogen peroxide and dioxymethylene, from which orthoformic acid is formed by further oxidation.

The substance obtained by Legler[199]

$$(CH_2O)_6O_3 + 3\ H_2O = (C_2H_6O_4)_3$$

arises from formaldehyde and hydrogen peroxide according to the equation

$$2\ H_2C{=}O + H{\cdot}O{\cdot}O{\cdot}H = \begin{matrix} OH & OH \\ H_2C{\cdot}O{\cdot}O{\cdot}CH_2 & \end{matrix};$$

it is therefore to be regarded as **formal hyperoxide hydrate**, completely analogous to the diphenylformal hyperoxide hydrate obtained from benzaldehyde.

That ethylene at lower temperature (120°) still burns exclusively to acetic acid (probably through intermediate uptake of water) is known from Berthelot's experiments[200]. Propylene, in similar fashion, gives with chromic acid at 50° only acetone. By contrast, ethylene burns with oxygen at 400°, according to Schützenberger[201], exclusively to formaldehyde — which is to be explained by a dissociation of the ethylene into two molecules of methylene.

The following experiments on the preparation of methylene are of interest in this connection. Methyl iodide, heated with mercury in air to 100°, is slowly converted into formaldehyde; the same is the case when the methylene addition products discovered by Sakurai[202], $CH_2{\cdot}HgJ_2$ and $CH_2{\cdot}Hg_2J_2$, are heated in air. If, on the other hand, one carries out the experiments in an atmosphere of nitrogen at 180°, only ethylene is formed; the primarily formed methylene, if it cannot oxidize to formaldehyde in air, thus simply polymerizes to ethylene.

k. The phosphorescence phenomena that occur in the slow oxidation of animals, plants, and carbon-containing bodies rest upon a dissociation into methylene derivatives, which then, quite analogously to phosphorus and other spontaneously combustible substances, burn in moist air with the same phenomena.

Schönbein, in his classic investigations on ozone and hydrogen peroxide, demonstrated that, exactly as in the burning of phosphorus, in oxidation processes in animals, plants, and carbon-containing compounds both active oxygen and hydrogen peroxide always appear. One important observation, however, here escaped…

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198) Millon, *Ann. chim. phys.* [3] **8**, 290; Klobukow, *Journ. f. pract. Chem.* [2] **34**, 126.

199) *Ber. d. deutsch. chem. Ges.* **18**, 3343.

200) *Compt. rend.* **68**, 334.

201) *Bull. soc. chim.* **31**, 482.

202) *Journ. chem. Soc.* **37**, 658; **39**, 486.

*English translation of: J. U. Nef, "Ueber das zweiwerthige Kohlenstoffatom. [Vierte Abhandlung.] Die Chemie des Methylens," Liebigs Annalen der Chemie **298** (1897), pp. 328–329 (excerpt).*